

NAN, Thailand

WMO: 483310

Lat: 18.7669N

Lon: 100.7633E

Elev: 661

StdP: 14.35

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	54.2	56.9	48.6	51.8	72.6	51.9	58.7	70.7	9.8	79.9	8.2	79.6	0.4	20	0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	23.2	100.8	76.8	98.8	76.7	96.8	76.7	81.7	91.7	81.0	90.8	80.5	90.0	2.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.4	156.4	84.9	78.9	153.6	84.5	78.3	150.5	84.1	46.2	92.1	45.4	91.1	44.8	90.5	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.2	6.9	49.4	104.4	3.5	2.1	46.9	105.8	44.8	107.0	42.8	108.2	40.2	109.7		
(5)			48.8	84.3	3.7	1.5	46.2	85.4	44.0	86.2	42.0	87.0	39.3	88.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.8	73.2	77.0	82.1	85.8	85.4	84.7	83.2	82.5	82.7	81.7	77.9	73.2
	DBStd	5.33	3.69	3.61	3.81	3.35	3.39	2.56	2.80	2.35	2.14	2.47	3.65	4.36
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	6	3	0	0	0	0	0	0	0	0	0	0	3
	CDD50	11238	720	757	994	1074	1096	1040	1031	1007	981	982	836	720
	CDD65	5769	257	337	529	624	631	590	566	542	531	517	386	259
	CDH74	59547	2161	3365	6003	7612	7332	6816	5992	5334	5160	4714	3094	1964
	CDH80	26701	828	1707	3444	4303	3607	3007	2302	1833	1876	1874	1237	683
(14) Wind	WSAvg	1.3	1.0	1.2	1.5	1.5	1.5	1.7	1.7	1.4	1.1	1.0	0.9	1.0
(15) Precipitation	PrecAvg	51.5	0.4	0.6	1.3	3.6	6.7	6.5	8.3	10.9	8.1	4.3	0.7	0.2
	PrecMax	66.0	2.6	2.8	3.7	6.9	13.4	10.9	17.1	16.9	16.3	9.0	2.7	1.5
	PrecMin	39.2	0.0	0.0	0.0	0.1	2.4	2.6	2.9	5.2	3.3	1.4	0.0	0.0
	PrecStd	7.0	0.7	0.8	1.0	2.2	2.7	2.4	3.5	3.2	4.2	2.0	0.8	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.8	97.3	102.2	104.3	102.5	98.5	96.8	94.1	94.3	94.1	93.5	91.2
		MCWB	72.1	72.7	75.2	78.0	79.2	79.5	78.8	80.4	80.0	78.4	76.7	74.6
	2%	DB	89.6	94.9	99.9	102.0	98.9	95.4	93.8	91.8	92.4	92.3	91.4	88.7
		MCWB	71.2	72.1	74.1	77.6	78.8	78.9	79.4	79.7	79.5	78.0	75.5	72.5
	5%	DB	87.6	92.9	97.8	99.4	96.5	93.4	91.7	90.0	90.8	90.9	89.5	86.5
		MCWB	70.2	71.7	74.0	77.5	78.9	78.9	79.1	79.2	79.1	77.6	74.9	71.4
	10%	DB	85.3	90.3	95.3	96.9	93.7	91.5	89.6	88.1	89.0	89.3	87.4	84.2
		MCWB	69.6	71.3	73.8	77.6	78.7	78.8	78.6	78.7	78.6	77.2	73.9	70.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.1	77.3	79.8	82.6	82.9	81.9	81.8	81.9	82.1	81.1	79.3	77.2
		MCDB	86.7	91.3	96.3	97.0	94.0	91.8	91.3	89.9	89.7	89.5	88.5	86.3
	2%	WB	73.9	75.4	77.9	80.7	81.3	81.0	80.8	80.9	81.0	80.0	78.1	75.1
		MCDB	84.3	89.6	93.6	95.0	92.7	90.6	90.2	88.8	88.7	88.5	87.0	83.7
	5%	WB	72.4	74.0	76.6	79.7	80.5	80.3	80.0	80.1	80.2	79.1	76.8	73.6
		MCDB	82.5	87.8	92.0	93.8	91.7	89.6	88.5	87.4	87.5	87.3	84.9	82.3
	10%	WB	71.1	72.6	75.4	78.7	79.7	79.7	79.2	79.4	79.5	78.3	75.7	72.2
		MCDB	81.2	85.7	90.4	92.8	90.1	88.3	86.8	86.0	86.3	85.9	83.0	80.5
(35) Mean Daily Temperature Range	5% DB	MDBR	26.5	29.1	27.8	23.2	17.5	14.4	12.9	12.5	14.0	16.6	20.3	23.6
		MCDBR	29.1	32.0	31.4	26.6	21.7	17.3	16.1	15.1	16.5	18.4	22.1	24.9
	5% WB	MCWBR	12.2	11.9	10.2	8.0	6.7	5.7	5.4	5.7	6.1	6.6	8.3	10.2
		MCWBR	25.8	28.5	27.4	23.6	18.7	15.4	14.4	14.0	15.1	16.8	19.5	22.1
(40) Clear-Sky Solar Irradiance	taub	taub	0.444	0.546	0.691	0.659	0.495	0.445	0.440	0.458	0.486	0.525	0.431	0.430
		taud	2.187	1.918	1.616	1.737	2.202	2.370	2.376	2.311	2.214	2.063	2.291	2.251
	Ebn at Noon	Ebn at Noon	263	241	212	221	258	269	270	268	260	244	266	264
		Edh at Noon	44	60	84	75	47	39	39	42	46	52	39	40
(44) All-Sky Solar Radiation	RadAvg	1413	1623	1690	1833	1676	1506	1386	1371	1448	1458	1468	1352	
	RadStd	82	78	101	122	151	115	112	90	69	74	85	78	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.72	N/A	N/A	N/A	N/A	+0.47	N/A	N/A	+277
(47) Regional (15 neighbors)		+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+265

Nomenclature: See separate page